

Inference Scaled GraphRAG: Improving Multi Hop Question Answering on Knowledge Graphs

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Abstract

Large Language Models (LLMs) have achieved impressive capabilities in language understanding and generation, yet they continue to underperform on knowledge intensive reasoning tasks due to limited access to structured context and multi-hop information. Retrieval-Augmented Generation (RAG) partially mitigates this by grounding generation in retrieved context, but conventional RAG and GraphRAG methods often fail to capture relational structure across nodes in knowledge graphs. We introduce Inference-Scaled GraphRAG, a novel framework that enhances LLM based graph reasoning by applying inference-time compute scaling.